

Interfacing of ADC with 8051

Resolution

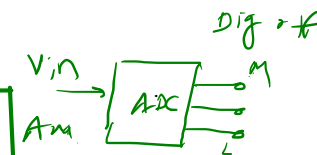
18

11

1.9

(Step size = 1V)

I/p Voltage (Analog)	Binary equivalent(Digital o/p)
0-1 V	000
1-2 V	001
2-3 V	010
3-4 V	011
4-5 V	100
5-6 V	101
6-7 V	110
7-8 V	111



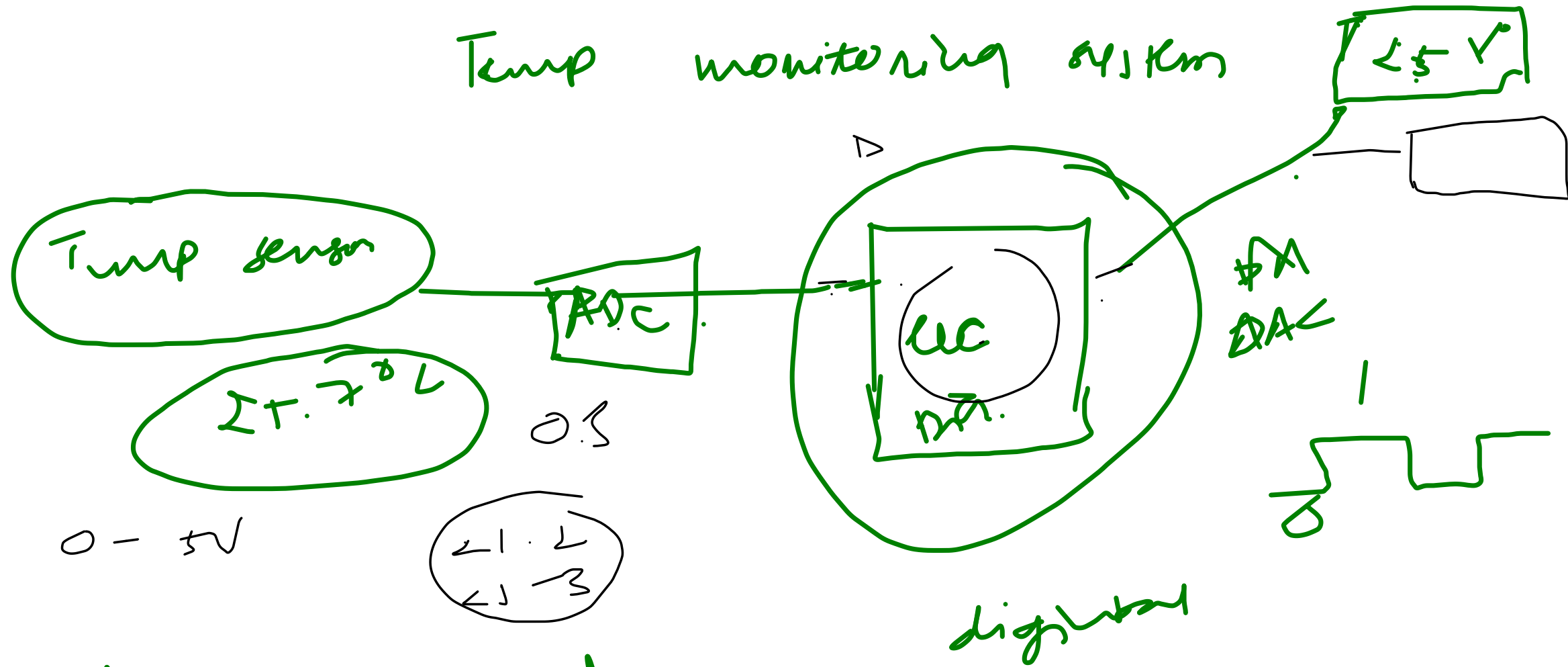
$$\text{Step Size} = \frac{V_{\text{max}} - V_{\text{min}}}{2^n} ; \quad n \equiv \text{no. of bits (Resolution)}$$

$V_m \rightarrow 2 \text{ to } 5$

$$\frac{5-2}{8} = \frac{3}{8} =$$

00
01
10
11

Temp monitoring system



Analog to Digital
converter
↓
Timer Varying

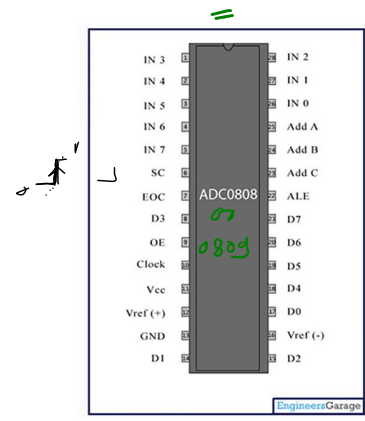
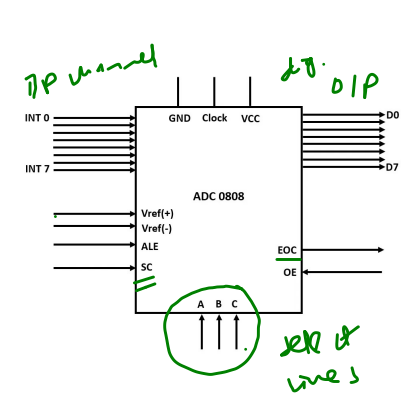
Number of bits	Number of steps	step size(mV)
<u>8</u>	256	5/256=19.53
10	1024	5/1024=4.88
12	4096	5/4096=1.2
16	65536	0.076 (precise conversion)

—

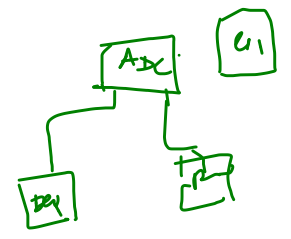
$V_{in} \rightarrow 0 \text{ to } 5V$

0 to 19.53 — 0.0000 —
5

NS
 IC chips: 0804 → single
0808/0809
 ↳ 8 Analog
 IP



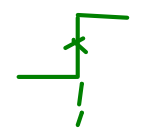
× serial ADC
 ✓ parallel ADC



Input pins (INT0-INT7) :

Select lines and ALE

0 C	B	0 A	Selected analog channel	ALE pin
0	0	0	INT0	Low to High pulse
0	0	1	INT1	Low to High pulse
0	1	0	INT2	Low to High pulse
0	1	1	INT3	Low to High pulse
1	0	0	INT4	Low to High pulse
1	0	1	INT5	Low to High pulse
1	1	0	INT6	Low to High pulse
1	1	1	INT7	Low to High pulse



Output pins (D0-D7):

VCC and Ground:

Clock:

10809

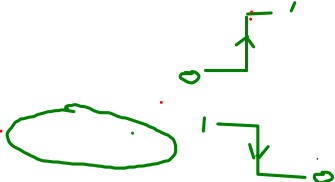
VREF (+) and VREF (-):

9

Start conversion



End of conversion



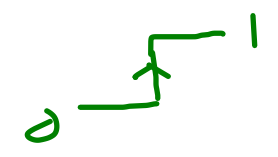
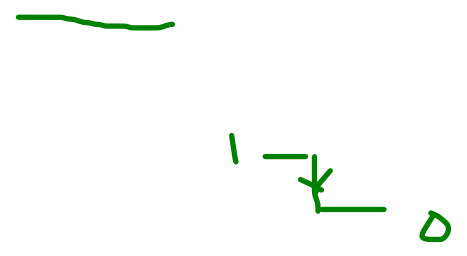
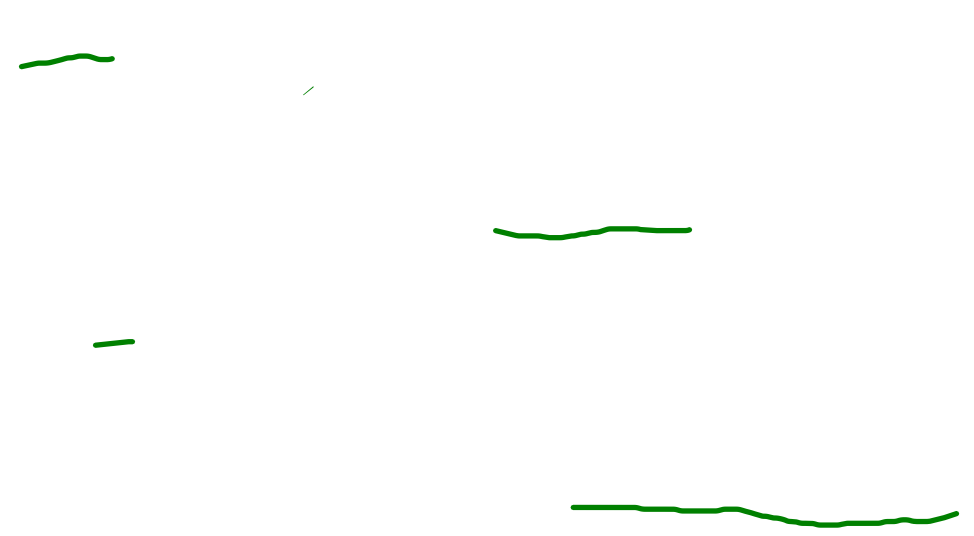
Output enable



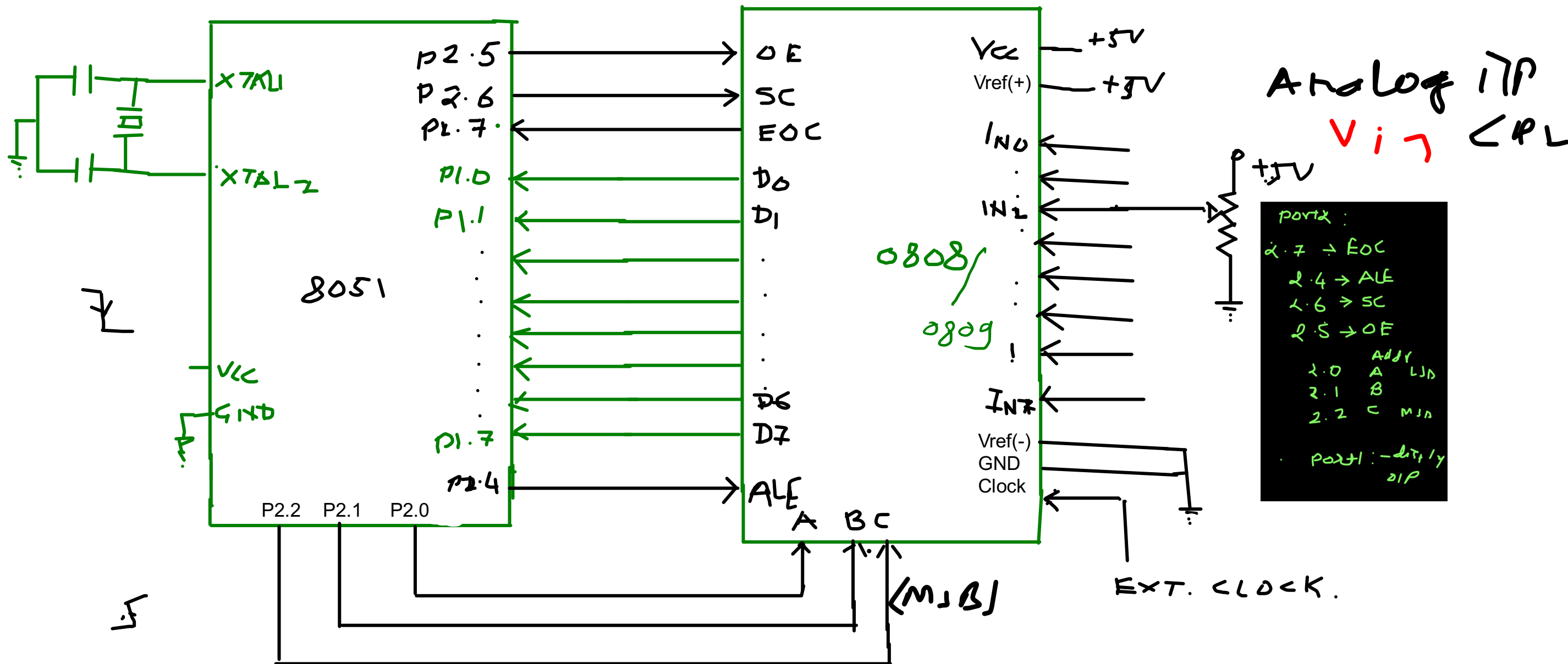
EOC $\overline{\Phi}$
Add IP to 8051

Port-2: Addn Lines ~~EB~~
SC $\begin{matrix} 0 & 1 & 0 \\ \downarrow & \downarrow & \downarrow \end{matrix}$
ALE $\downarrow$

1080g:



Interfacing 8051 with ADC-0808/0809



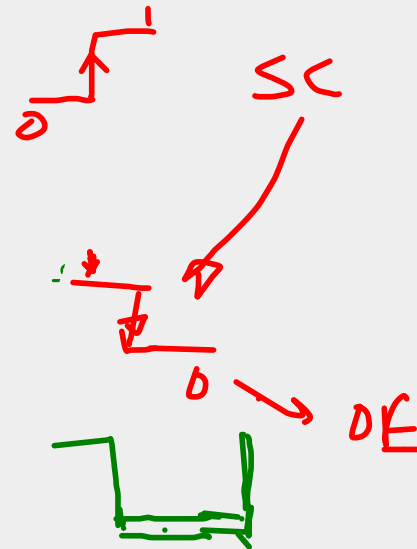
Port 1: Digital O/P ; Port 2 → To control ADC

```

ORG 0000H; Starting address
MOV P1,#0FFH; Makes port 1 input port
SETB P2.7; Makes EOC pin high
CLR P2.4; Clears ALE pin
CLR P2.6; Clears Start pin
CLR P2.5; Clears OE pin
BACK: CLR P2.2; Clears ADD C
      SETB P2.1; Sets ADD B,
      CLR P2.0; Clears ADD A (this selects the second address line)
      ACALL DELAY
      SETB P2.4; Sets ALE high
      ACALL DELAY
      SETB P2.6; sends a command to start conversion
      ACALL DELAY
      CLR P2.4; makes ALE low
      CLR P2.6; makes Start pin low
      HERE: JB P2.7,HERE; waits for low pulse at EOC
      HERE1: JNB P2.7,HERE1; waits for low pulse to finish
      SETB P2.5; enables OE pin to extract data from ADC
      ACALL DELAY
      MOV A,P1; moves acquired data to accumulator
      CLR P2.5; clears OE
      SJMP BACK; repeatedly gets data from ADC
      DELAY: MOV R3,#50
      HERE2: MOV R4,#255
      HERE3: DJNZ R4,HERE3
      DJNZ R3,HERE2
      RET
      END

```

C B A
 0 1 0
 IN 2

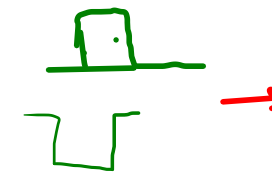


port 2 :

2.7 → EOC
 2.4 → ALE
 2.6 → SC
 2.5 → OE

Addr	
2.0	A LJD
2.1	B
2.2	C MIN

port 1 : - digital
o/p



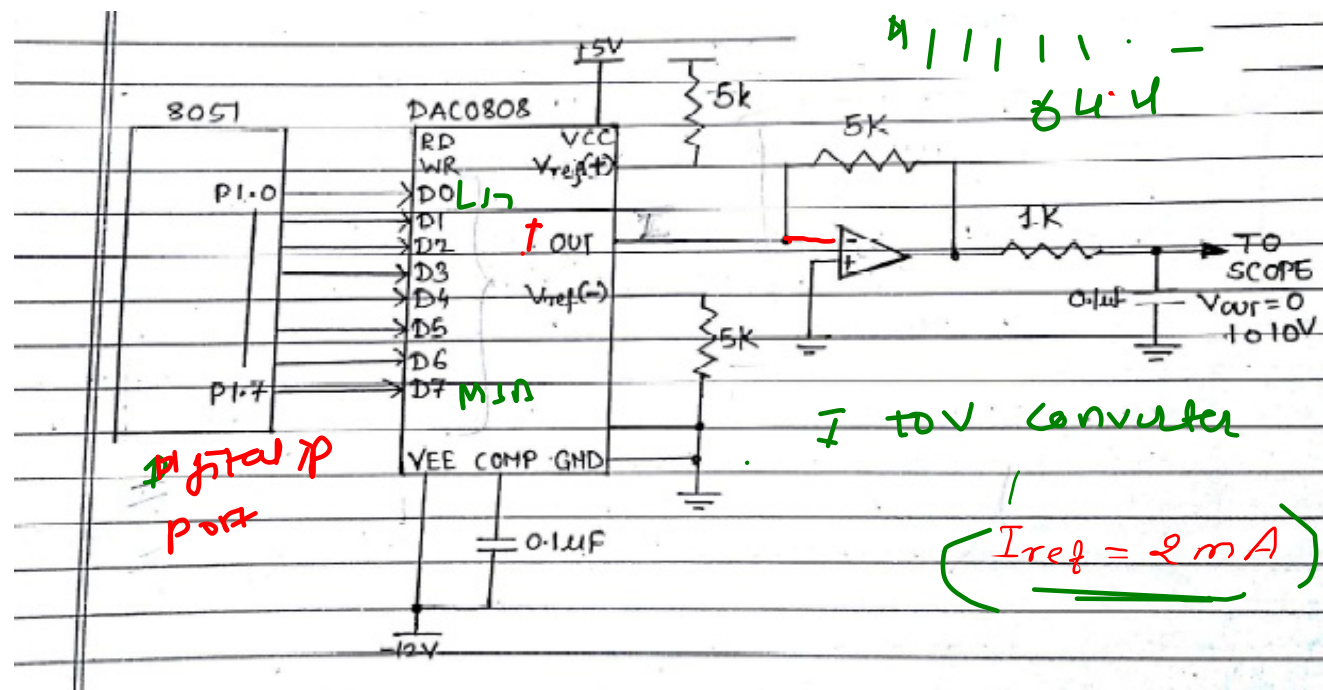
• DAC INTERFACING

DAC0808

In DAC0808, the digital inputs are converted to current (I_{out}), and by connecting a resistor to the I_{out} pin, we convert the result to voltage. The total current provided by the I_{out} pin is a function of the binary numbers at the D0-D7 inputs of the DAC0808 & the reference current (I_{ref}) & is as follows

$$I_{out} = I_{ref} \left(\frac{D7}{2} + \frac{D6}{4} + \frac{D5}{8} + \frac{D4}{16} + \frac{D3}{32} + \frac{D2}{64} + \frac{D1}{128} + \frac{D0}{256} \right)$$

where, D0 is the LSB, D7 is the MSB for the inputs & I_{ref} is the input current that must be applied to pin 14.



$$I_{out} = 2mA \left(\frac{0}{2} + \frac{1}{4} + \frac{1}{8} + \frac{0}{16} + \frac{0}{32} + \frac{1}{64} + \frac{1}{128} + \frac{0}{256} \right)$$

$$2mA \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{64} \right)$$

$$I_{ref} = 2mA$$

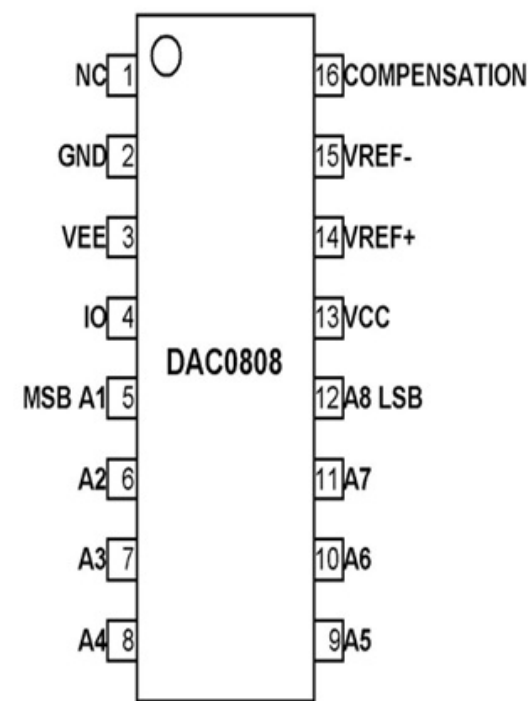
$$V_{out} = I_{out} \cdot 5k$$

$$V_{out} =$$

for bit IP
RFH

I_{out} =

DAC 0808 Pin diagram:



DAC0808 is a sixteen pin device and description for each pin is given below.

Pin	Name	Description
1	NC	No connection
2	GND	Ground
3	VEE	Negative power supply
4	IO	Output signal pin
5	A1	Digital input bit 1 (Most Significant Bit)
6	A2	Digital input bit 2
7	A3	Digital input bit 3
8	A4	Digital input bit 4
9	A5	Digital input bit 5
10	A6	Digital input bit 6
11	A7	Digital input bit 7
12	A8	Digital input bit 8 (Least Significant Bit)
13	VCC	Positive power supply
14	VREF+	Positive reference voltage
15	VREF-	Negative reference voltage
16	COMPENSATION	Compensation capacitor pin

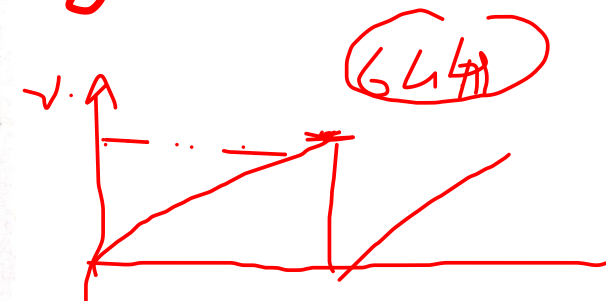
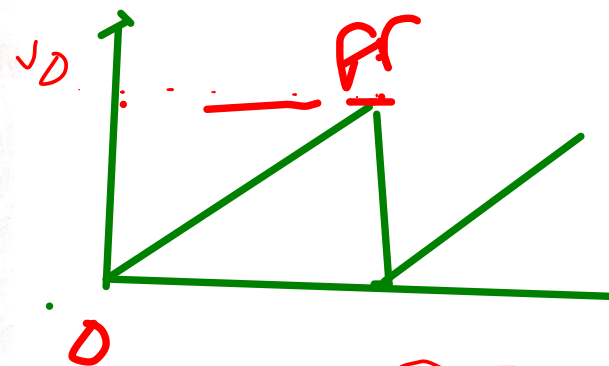
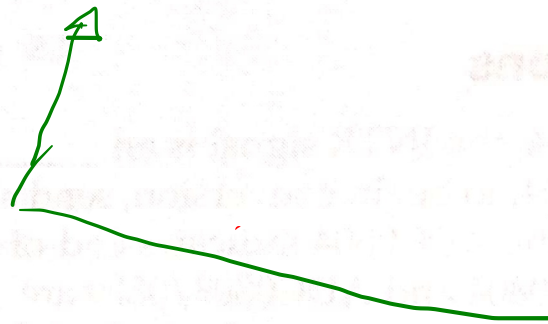
For the circuit of Figure 13-18, find the maximum output amplitude of the saw tooth waveform obtained with the following program.

(a) ~~Back:~~ MOV A, #00H
 MOV P1, A
~~BACK:~~ INC A
 SJMP BACK

(b) MOV R0, #64H
 RPT: MOV A, #00H
 BACK: MOV P1, A
 INC A
 CJNE A, R0, BACK
 SJMP RPT

Solution:

(a) Here, the method to generate the saw tooth wave is to increment the A register from 0 to FF to 0 continuously. In this case, the maximum value of the digital number is FFH = 11111111B.



$$I_{\text{out}} = I_{\text{ref}} \left(\frac{D7}{2} + \frac{D6}{4} + \frac{D5}{8} + \frac{D4}{16} + \frac{D3}{32} + \frac{D2}{64} + \frac{D1}{128} + \frac{D0}{256} \right)$$

$$= 2 \text{ mA} \times 255/256 = 2 \times 0.996 = 1.99 \text{ mA.}$$

$$V_{\text{out}} = 1.99 \text{ mA} \times 5 \text{ K} = 9.96 \text{ V}$$

The peak amplitude of the saw tooth waveform is 9.96 V.

(b) In this case, the maximum value of the digital number is 64 H = 01100100 B. With the same calculation,

$$I_{\text{out}} = 0.78125 \text{ mA}$$

$$V_{\text{out}} = 3.906 \text{ V}$$

$$2\text{mA} \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} \right) = 0.78125\text{mA}$$

Interfacing LCD with 8051:

LCD INTERFACING			
LCD pin descriptions			
The LCD has 14 pins. The function of each pin is given in the following table.			
Pin	Symbol	I/O	Description
1	Vss	—	Ground
2	Vcc	—	+5V power supply
3	Vee	—	Power supply to control contrast.
4	RS	I	RS=0 to select command register RS=1 to select data register
5	R/W	I	R/W=0 for write, R/W=1 for read
6	E	I/O	Enable
7	DB0	I/O	The 8-bit data bus
8	DB1	I/O	The 8-bit data bus
9	DB2	I/O	The 8-bit data bus
10	DB3	I/O	The 8-bit data bus
11	DB4	I/O	The 8-bit data bus
12	DB5	I/O	The 8-bit data bus
13	DB6	I/O	The 8-bit data bus
14	DB7	I/O	The 8-bit data bus

1) Vcc, Vss, and Vee

While Vcc and Vss provide +5V and ground, respectively, Vee is used for controlling LCD contrast.

2) RS, register select

If RS=0, the instruction/command code register is

selected, allowing the user to send a command such as clear display, cursor at home etc.

If RS=1, the data register is selected, allowing the user to send data to be displayed on the LCD.

3) R/W, read/write

R/W input allows the user to write information to the LCD or read information from it.

R/W=1 when reading; R/W=0 when writing.

4) E, enable

The enable pin is used by the LCD to latch information presented to its data pins.

When data is supplied to data pins, a high-to-low pulse must be applied to this pin in order for the LCD to latch in the data present at the data pins.

5) D0-D7

The 8-bit data pins, D0-D7, are used to send information to the LCD or read the contents of the LCD's internal registers.

To display letters and numbers, we send ASCII codes for the letters A-Z, a-z to number 0-9 to these pins while making RS=1.

We use RS=0 to check the busy flag bit to see if the LCD is ready to receive information. The busy flag is D7 & can be read when R/W=1.

When D7=1, the LCD is busy taking care of internal operations & will not accept any new information.

When D7=0, the LCD is ready to receive new information.

Interfacing LCD with 8051:

LCD INTERFACING			
LCD pin descriptions			
The LCD has 14 pins. The function of each pin is given in the following table.			
Pin	Symbol	I/O	Description
1	Vss	—	Ground
2	Vcc	—	+5V power supply
3	Vee	—	Power supply to control contrast.
4	RS	I	RS=0 to select command register RS=1 to select data register
5	R/W	I	R/W=0 for write, R/W=1 for read
6	E	I/O	Enable
7	DB0	I/O	The 8-bit data bus
8	DB1	I/O	The 8-bit data bus
9	DB2	I/O	The 8-bit data bus
10	DB3	I/O	The 8-bit data bus
11	DB4	I/O	The 8-bit data bus
12	DB5	I/O	The 8-bit data bus
13	DB6	I/O	The 8-bit data bus
14	DB7	I/O	The 8-bit data bus

1) Vcc, Vss, and Vee

While Vcc and Vss provide +5V and ground, respectively, Vee is used for controlling LCD contrast.

2) RS, register select

If RS=0, the instruction/command code register is

selected, allowing the user to send a command such as clear display, cursor at home etc.

If RS=1, the data register is selected, allowing the user to send data to be displayed on the LCD.

3) R/W, read/write

R/W input allows the user to write information to the LCD or read information from it.

R/W=1 when reading; R/W=0 when writing.

4) E, enable

The enable pin is used by the LCD to latch information presented to its data pins.

When data is supplied to data pins, a high-to-low pulse must be applied to this pin in order for the LCD to latch in the data present at the data pins.

5) D0-D7

The 8-bit data pins, D0-D7, are used to send information to the LCD or read the contents of the LCD's internal registers.

To display letters and numbers, we send ASCII codes for the letters A-Z, a-z to number 0-9 to these pins while making RS=1.

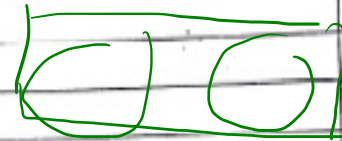
We use RS=0 to check the busy flag bit to see if the LCD is ready to receive information. The busy flag is D7 & can be read when R/W=1.

When D7=1, the LCD is busy taking care of internal operations & will not accept any new information.

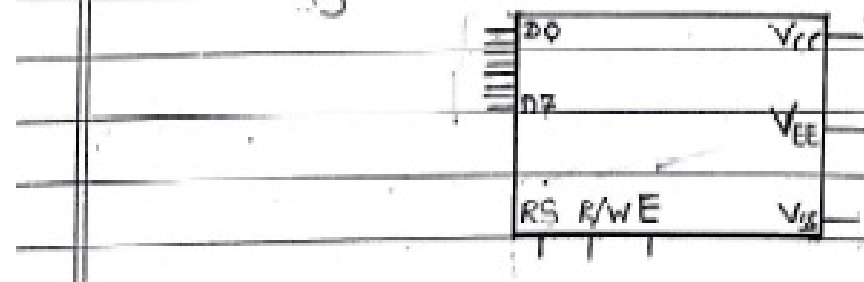
When D7=0, the LCD is ready to receive new information.

• LCD Command Codes

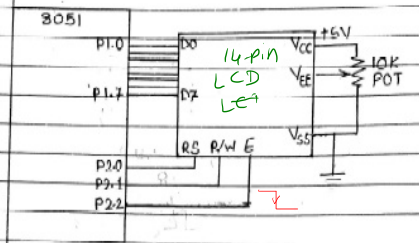
Code (Hex)	Command to LCD Instruction Register
01	Clear display screen
02	Return home
04	Decrement cursor (shift cursor to left)
06	Increment cursor (shift cursor to right)
05	Shift display right
07	Shift display left
08	Display off, cursor off
0A	Display off, cursor on
0C	Display on, cursor off.
0E	Display on, cursor blinking <i>on</i>
0F	Display <i>on</i> , cursor blinking <i>25/2</i>
10	Shift cursor position to left
14	Shift cursor position to right
18	Shift the entire display to the left.
1C	Shift the entire display to the right.
80	Force cursor to beginning of 1 st line
C0	Force cursor to beginning of 2 nd line.
38	2 lines and 5x7 matrix.



• Pin Diagram of LCD



Interfacing of LCD with 8051



To send any of the commands to the LCD, make pin RS=0. For data, make RS=1. Then send a high-to-low pulse to the E pin to enable the internal latch of the LCD. This is shown in following program

```

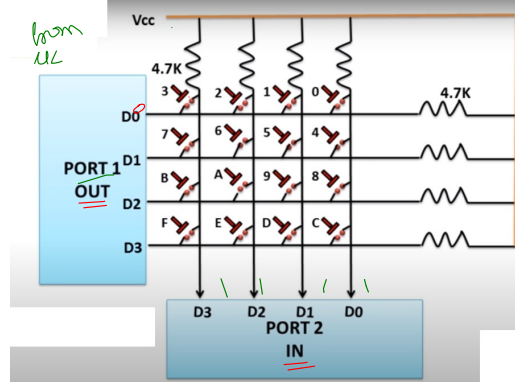
ORG 0H
MOV A, #38H ; init LCD 2 lines, 5x7 matrix.
ACALL COMNWRT ; call command subroutine
ACALL DELAY ; give LCD some time
MOV A, #0EH ; display on cursor on
ACALL COMNWRT ; call command subroutine
ACALL DELAY ; give LCD some time
MOV A, #01 ; clear LCD
ACALL COMNWRT ; call command subroutine
ACALL DELAY ; give LCD some time

MOV A, #0EH ; shift cursor right
ACALL COMNWRT ; call command subroutine
ACALL DELAY ; give LCD some time
MOV A, #84H ; cursor at line 1, pos. 4
ACALL COMNWRT ; call command subroutine
ACALL DELAY ; give LCD some time
MOV A, #'H' ; display letter H
ACALL DATAWRT ; call display subroutine
ACALL DELAY ; give LCD some time
MOV A, #'0' ; display letter 0
ACALL DATAWRT ; call display subroutine
AGAIN: SJMP AGAIN ; stay here

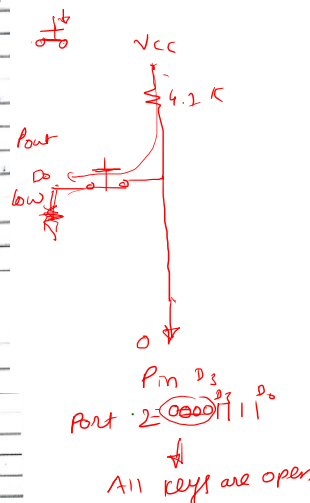
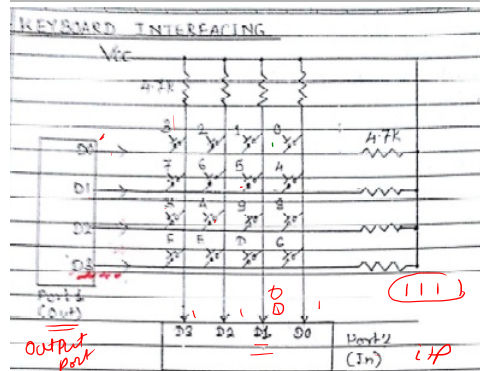
COMNWRT: ; send command to LCD
MOV P1, A ; copy reg A to port 1
CLR P2.0 ; RS=0 for command
CLR P2.1 ; R/W=0 for write
SETB P2.2 ; E=1 for high pulse
ACALL DELAY ; give LCD some time
CLR P2.2 ; E=0 for H-to-L pulse
RET

DATAWRT: ; write data to LCD
MOV P1, A ; copy reg A to port 1
SETB P2.0 ; RS=1 for data
CLR P2.1 ; R/W=0 for write
SETB P2.2 ; E=1 for high pulse
ACALL DELAY ; give LCD some time
CLR P2.2 ; E=0 for H-to-L pulse
RET

DELAY: MOV R3, #50 ; 50 or higher for fast CPUs
HERE?: MOV R4, #255 ; R4 = 255
HERE: DJNZ R4, HERE ; stay until R4 becomes 0
DJNZ R3, HERE?
RET
END
    
```



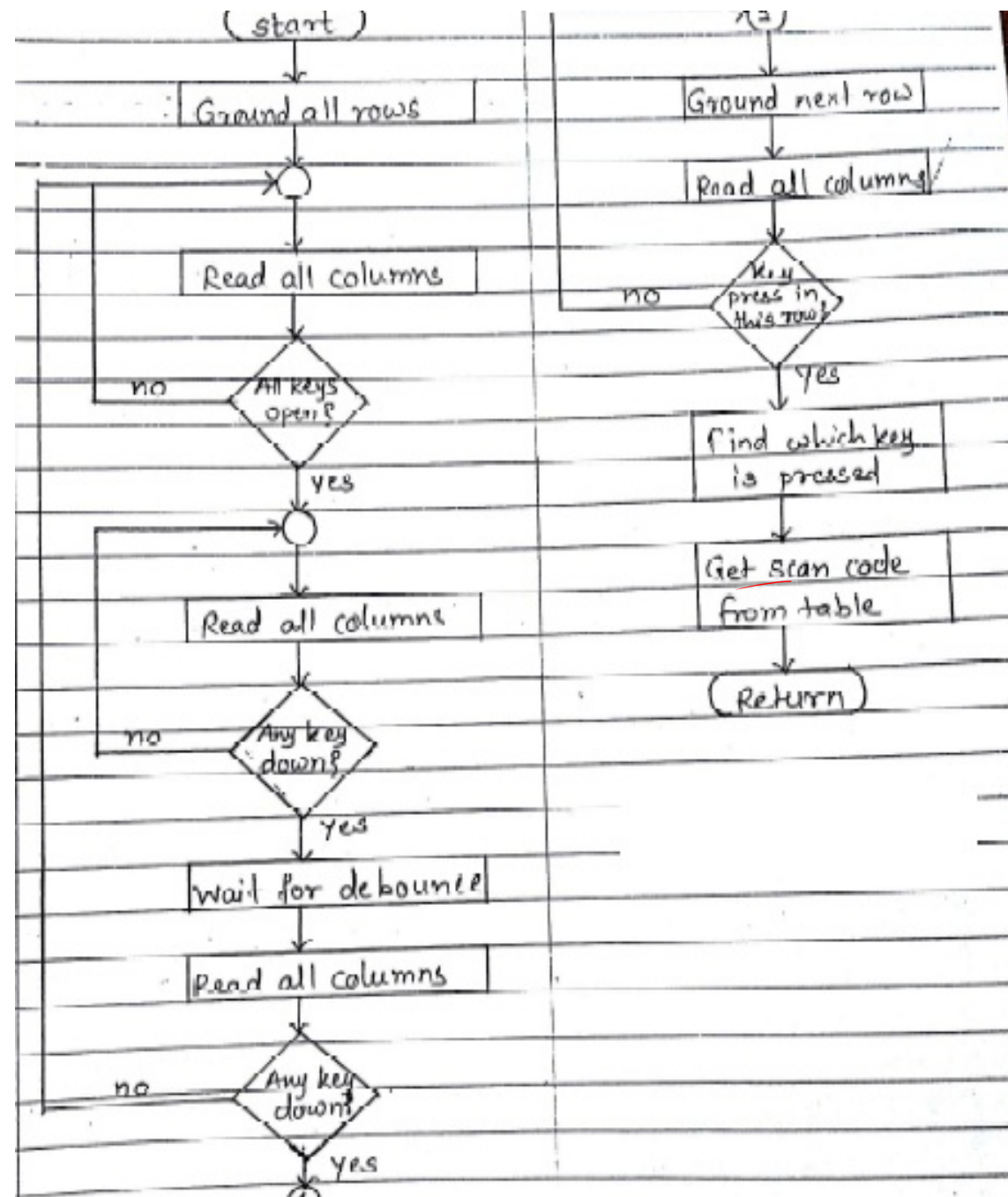
4x4 matrix
key board



Scanning and identifying the key
 1) Above figure shows a 4x4 matrix connected to two ports.
 2) The rows are connected to an output port and the columns are connected to an input port.
 3) If no key has been pressed, then pins of port 2 will be 1 since they are all connected to Vcc.
 4) If all the rows are grounded and a key is pressed, one of the columns i.e. one of the pins of port 2 will have 0 since the key pressed provides the path to ground.
 5) It is the function of the microcontroller to scan the keyboard continuously to detect and identify the key pressed.

6) To detect a pressed key, the microcontroller grounds all rows by providing 0 to the output latch, then it reads the columns.
 • If the data read from columns i.e. in port 2 from D3 to D0 is 1111, no key has been pressed and the process continues till key press is detected.
 • If one of the column bits has a zero, this means that a key press has occurred. e.g. if from D3 to D0 we have 1101, this means that a key in the D1 column has been pressed.
 7) After detecting a key press, microcontroller will go through the process of identifying the key.
 8) Starting with the top row, the microcontroller grounds it by providing a low to row D0 only.
 • It reads the columns, if the data read is all 1s i.e. in port 2 from D3 to D0 is 1111, no key in that row is activated and the process is moved to the next row.
 9) It grounds the next row, reads the columns, and checks for any zero. This process continues until the row is identified.
 10) After identification of the row in which the key has been pressed, find out which column the pressed key belongs to.

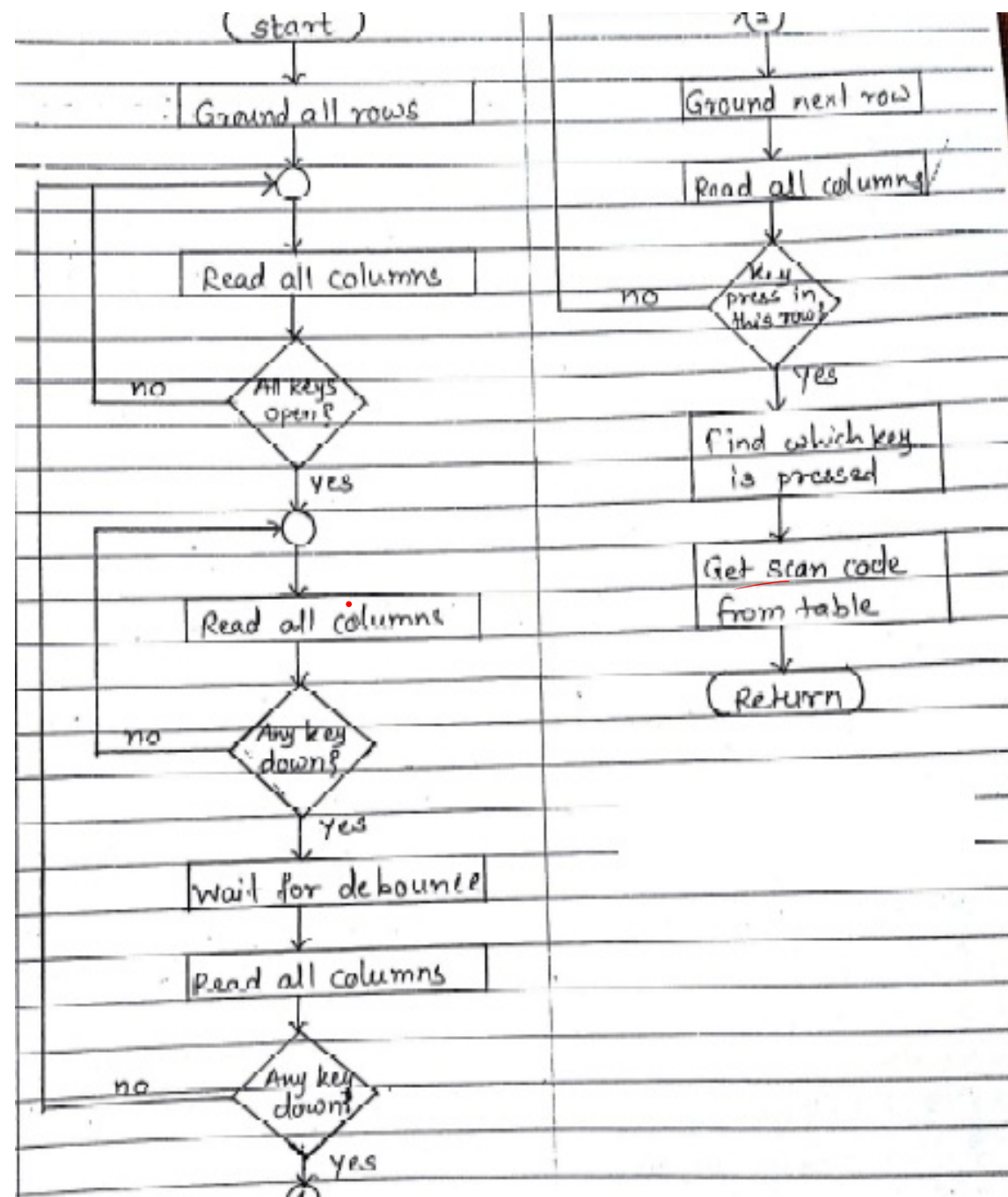
0	1	2	3
---	---	---	---

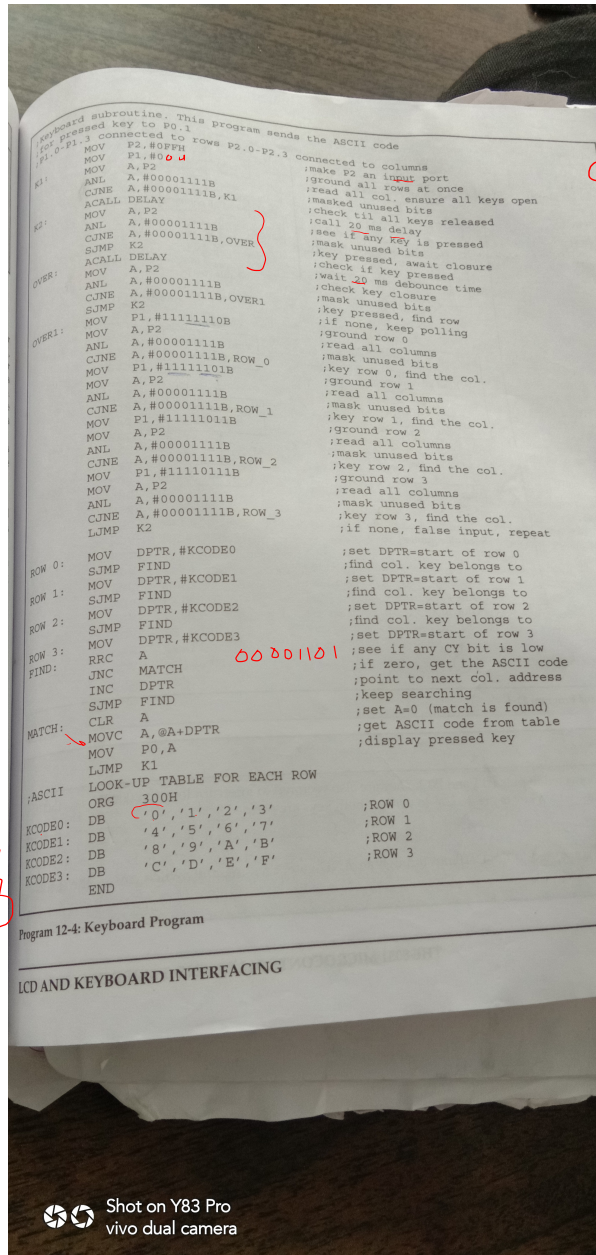


P₂

1
0 =

^





- - D3 D2 D1 D0
- 1 1 1 0
A = 0000 1110
RRC A
CY

D4 P5 2, D0
0000 1 0 1 1
0000 1 1 1 1
0000 1 0 1 1
P2
1 1 1 1
3 2 2 1 P2.0
P2 = 0000 1111
and 0000 1111
0000 1111
2 3 2 2 2 1 2 0

